FINITE TYPE DESCENDS ALONG A SURJECTIVE FLAT OPEN MORPHISM OF LOCALLY RINGED SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Although the label speaks of fppf descent, the statement is made for an arbitrary morphism of locally ringed spaces that is open, surjective and flat; the ledger entry follows the label, the card follows the statement.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A flat local ring map is faithfully flat. Algebra, Lemma algebra-lemma-local-flat-ff, invoked at proof lines 2162-2164. Upgrades surjectivity of the stalk map after tensoring to surjectivity of the stalk map itself, which is the only point where the flatness hypothesis is used.
- The pullback of a sheaf of modules as a tensor product of the inverse image sheaf with the structure sheaf. Standard, used at proof lines 2143-2146 to rewrite generators of the pullback as sums of tensors. Allows the chosen generators of the pulled back sheaf to be expressed through sections of the original sheaf.
- Finite type for a sheaf of modules, checked on stalks and localised on an open neighbourhood. Standard, used at proof lines 2141-2142 and again at proof lines 2165-2166. Reduces the global assertion to producing finitely many generators in a neighbourhood of each point of the target.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of locally ringed spaces. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_Y$-modules. If*

(1) *f is open as a map of topological spaces,*
(2) *f is surjective and flat, and*
(3) *$f^*\mathcal{F}$ is of finite type,*

then $\mathcal{F}$ is of finite type.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $y \in Y$ be a point. Choose a point $x \in X$ mapping to y. Choose an open $x \in U \subset X$ and elements $s_1, \ldots, s_n$ of $f^*\mathcal{F}(U)$ which generate $f^*\mathcal{F}$ over U. Since $f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$ we can after shrinking U assume $s_i = \sum t_{ij} \otimes a_{ij}$ with $t_{ij} \in f^{-1}\mathcal{F}(U)$ and $a_{ij} \in \mathcal{O}_X(U)$. After shrinking U further we may assume that t_{ij} comes from a section $s_{ij} \in \mathcal{F}(V)$ for some $V \subset Y$ open with $f(U) \subset V$. Let N be the number of sections s_{ij} and consider the map

$$\sigma = (s_{ij}) : \mathcal{O}_V^{\oplus N} \to \mathcal{F}|_V$$

By our choice of the sections we see that $f^*\sigma|_U$ is surjective. Hence for every $u \in U$ the map

$$\sigma_{f(u)} \otimes_{\mathcal{O}_{Y,f(u)}} \mathcal{O}_{X,u} : \mathcal{O}_{X,u}^{\oplus N} \longrightarrow \mathcal{F}_{f(u)} \otimes_{\mathcal{O}_{Y,f(u)}} \mathcal{O}_{X,u}$$

is surjective. As f is flat, the local ring map $\mathcal{O}_{Y,f(u)} \to \mathcal{O}_{X,u}$ is flat, hence faithfully flat (Algebra, Lemma 00HR). Hence $\sigma_{f(u)}$ is surjective. Since f is open, $f(U)$ is an open neighbourhood of y and the proof is done. $\square$